

Cryptography and Network Security

Contents

- Outline
 - Zero-knowledge proof
 - Fiat-Shamir protocol

Zero Knowledge Proofs

A **Zero-Knowledge Proof** is a cryptographic method (protocol) where:

A **Prover** wants to prove something is true.

A **Verifier** wants to be convinced.

The Prover proves the statement **without revealing any secret information**.

“Prove that you know the secret — without revealing the secret.”

Examples:

Prove that a number $n = p \times q$ (product of two primes) $\rightarrow$ Without revealing p and q

Prove that one knows x such $y = g^x \bmod p$ $\rightarrow$ Without revealing x

Two Kinds of Zero-Knowledge Proofs

ZK proof of a statement

- convincing the verifier that a **statement is true** without revealing any other information
- Example: Proving that a propositional formula is satisfiable (i.e., a solution exists) → Without revealing the actual solution.

ZK proof of knowledge

- convincing the verifier that the prover **knows a secret** → Without revealing the secret itself
- Example: Proving knowledge of x such that $y = g^x \bmod p$ → Without revealing x

Properties of Interactive Zero-Knowledge Proofs of Knowledge

Completeness

- Given **honest prover and honest verifier**, the **protocol succeeds with overwhelming probability**

Soundness

- **A prover who does not know the secret cannot convince the verifier (except with a very negligible probability).**

Zero knowledge

- The proof **does not leak any additional information**

Fiat-Shamir Protocol for Proving Quadratic Residues

Statement: x is QR modulo n

Prover knows w such that $w^2 \equiv x \pmod{n}$

The prover wants to prove the knowledge of w without revealing w itself.

Repeat the following one-round protocol t times for higher security:

One-round Protocol:

- **Commitment ($P \rightarrow V$):** Choose random r ; send $y = r^2 \pmod{n}$
- **Challenge ($V \rightarrow P$):** Verifier sends random bit $b \in \{0,1\}$
- **Response ($P \rightarrow V$):** Send $z = rw^b$, i.e., $z=r$ if $b=0$, $z=rw$ if $b=1$
- **Verification (V checks):** Check: $z^2 \equiv y \cdot x^b \pmod{n}$, i.e., $z^2=y$ if $b=0$, $z^2=yx$ if $b=1$ [Because $z=rw^b$ and therefore $z^2 = r^2 (w^b)^2 = r^2 (w^2)^b$; Now $y = r^2 \pmod{n}$ and $w^2 \equiv x \pmod{n}$; hence $z^2 \equiv y \cdot x^b \pmod{n}$]

The prover proves the knowledge of w without revealing w itself.

Example

Let: $n=11$ secret $w=4$

Compute: $w^2=16 \equiv 5 \pmod{11}$

So public values are: $n=11$ $x=5$

One Round Execution

1. Commitment ($P \rightarrow V$)

Choose random $r=3$

$$y=r^2=9 \pmod{11}$$

2. Challenge ($V \rightarrow P$)

Choose random bit $b=1$

...Example

3. Response ($P \rightarrow V$)

$$z = r \cdot w^b = 3 \cdot 4 = 12 \equiv 1 \pmod{11}$$

4. Verification (V checks)

$$\text{Check: } z^2 \equiv y \cdot x^b \pmod{11}$$

$$\text{Left: } 1^2 = 1 \quad \text{AND} \quad \text{Right: } 9 \cdot 5 = 45 \equiv 1 \pmod{11}$$

✓ Verified

Observations on the Protocol

Multiple rounds

Each round consists of 3 steps

- Commit; Challenge; Respond

If challenge can be predicted, then cheating is possible.

If **respond to more than one challenge with one commit**, then the secret is revealed (because b has only 2 possible values: 0 and 1).

Analysis of the Fiat-Shamir protocol

1. Completeness

If the prover truly knows w such that $w^2 \equiv x \pmod{n}$ and both parties follow the protocol correctly, the verifier **always accepts**.

2. Soundness

If x is **NOT** a **quadratic residue**, a cheating prover cannot always fool the verifier.

* For any commitment y , the prover can prepare for only **one** challenge ($b = 0$ or $b = 1$), not both.

* Since the verifier chooses $b \in \{0,1\}$ randomly, the cheating prover succeeds with probability at most: $\frac{1}{2}$. After repeating the protocol t **times**, cheating success probability becomes: $(1/2)^t$

- Assume that x is not QR, V receives y
 - Case 1: y is QR, then when $b=1$, checking $z^2=yx$ will fail.
 - Case 2: y is QNR, then when $b=0$, checking $z^2=y$ will fail.

What if challenge is predicted?

Problem: Suppose the prover **knows the challenge bit b** in advance. Then the prover can cheat by preparing the commitment accordingly.

Cheating Strategy: If the prover predicts challenge b , he can choose any value z and compute commitment as: $y = z^2 x^{-b} \bmod n$ (instead of $y = r^2 \bmod n$)

During Verification

Verifier checks: $z^2 = yx^b$

Substitute y : $z^2 = (z^2 x^{-b})x^b = z^2$

✓ Verification always succeeds!

If the challenge is predictable: The prover can prepare a fake proof and hence, Soundness breaks

Exercise

Given: $n = 13$, $x = 9$, Commitment: $y = 3$, Challenge: $b = 1$, Response: $z = 5$
Verify whether the proof is valid

For the same commitment y , the verifier gets: For $b = 0$: $z_0 = 4$ and
For $b = 1$: $z_1 = 8$ $n = 11$ Find the secret w

Given: $n = 11$, $x = 5$ Prover predicts $b = 1$ Chooses $z = 3$
Find the commitment y such that verification passes